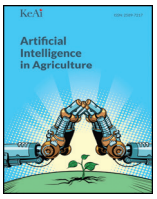
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PVDT: A smart tool for rapid detection of pollen viability

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ABSTRACT

Pollen viability is a critical determinant of maize reproductive capacity and genetic potential, directly influencing pollination efficiency, seed set, and overall crop yield. While traditional detection methods are labor-intensive and time-consuming, rapid alternatives like flow cytometry remain prohibitively expensive, and existing machine vision approaches lack specialized tools. To address these challenges, the study introduces "AIpollen" software and a cost-effective Pollen Viability Detection Tool (PVDT) device. The software employs a dual-pathway detection system: (1) the YOLOv10-CBAM model, which achieves 90%–98% accuracy even under challenging conditions (e.g., variable lighting, high pollen density and low image quality), outperforming existing advanced models; and (2) the Pollen Microscopic Image Segmentation Algorithm (PMISA). When combined with a Residual Dual-Attention Gated Network (RDAGNet) model, PMISA mitigates performance degradation in high-density or low-quality images, boosting detection accuracy by approximately 7%. Field trials utilizing the PVDT evaluated pollen viability in response to heat stress for 350 maize inbred lines, supporting the exploration of loci associated with pollen viability through GWAS. Furthermore, Transfer learning enabled the detection of pollen viability across ten additional plant species (e.g., wheat, peas, alfalfa, potato, tobacco) with favorable detection performance.

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1. Introduction

Maize (*Zea mays* L.) is the leading cereal crop in terms of worldwide production, with significant applications in human food, animal feed and industrial resources (Erenstein et al., 2022). During reproductive growth, pollen viability is a critical indicator of reproductive capacity, which directly determines seed set rates and grain yield in maize (Althiab-Almasaud et al., 2024). However, maize pollen viability was highly sensitive to high temperature (Rieu et al., 2017), and hot days and heat waves are predicted to increase in frequency and intensity in the coming decades (IPCC, 2022). Thus, attracting attention to pollen viability research is crucial for achieving high yield in maize and maintaining food security in the future, while efficient and accurate evaluation of pollen viability is a prerequisite for such researches.

For a long time, traditional pollen viability detection methods, such as TTC (2,3,5-triphenyltetrazolium chloride) staining method, acetocarmine staining method, I_2-KI staining method, and in vitro pollen germination test, could reflect the activity level of pollen to a certain extent. However, they generally have limitations such as cumbersome operation, time-consuming processes, and strong subjectivity in interpreting results (Olsson et al., 2021). Currently, rapid pollen viability detection technologies have been developed basing on flow cytometry technology and computer-aided image analysis (Mao et al., 2025). Nevertheless, these rapid methods are generally high-cost, which many researchers cannot be afforded.

Recently, target detection technology significantly penetrated agricultures, with applications primarily concentrated in pest and disease identification (John et al., 2023), weed monitoring (Gupta et al., 2023), and agricultural product yield assessment (Solimani et al., 2024). However, its application in the more elaborate pollen viability detection, is inadequate, primarily due to the high professional threshold of pollen viability assessment. That is, this assessment requires the deep integration of profound foundations in agronomy with advanced technologies in computer science, which strongly demands interdisciplinary expertise. With the development of deep learning technologies, the emergence of a series of object detection networks represented by RCNN and YOLO has injected strong impetus into the intelligent transformation of agriculture. For example, researchers utilized the YOLOv8

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network to detect the emergence rate of cabbage in drone imagery with an accuracy rate of 90.3% (Chen et al., 2024). Similarly, others employed the Faster R-CNN network to detect the number of wheat leaves in ground-based phenotypic platforms, achieving with an R^2 of the 0.94 (Li et al., 2023b). Furthermore, introducing attention mechanisms has been widely verified to effectively improve the generalization and robustness of YOLO-based detection models. For instance, integrating Convolutional Block Attention Module (CBAM), Spatial Attention (SA), incremental learning with domain adaption can significantly enhance detection accuracy for various plant organs in tomatoes (Cardellicchio et al., 2025). Those technological progress offers a new method for the direct detection of pollen viability.

Nevertheless, a challenge exists in applying target detection models in agriculture: complex image backgrounds significantly impacted the output results (Zhu et al., 2025). This is primarily due to the presence of interfering elements in the background that often resemble the target objects. Furthermore, high-density crop scenarios also pose significant challenges to detection model performance (Du and Liu, 2023; Li et al., 2023a), leading to a notable decline in model accuracy. Additionally, the mutual occlusion and overlap among targets make it challenging to accurately separate and identify them (Gui et al., 2024). Specifically, concerning to maize pollen detection, a large amount of pollen can lead to dense overlapping in the image, which increases recognition difficulty and introduces significant noise information. To overcome the performance limitations of object detection algorithms under high-density and complex background conditions, researchers widely adopted a two-stage approach combining segmentation with classification (Pang et al., 2025). This is because segmentation algorithms enable precise target-background separation; when integrated with classification models for pollen viability assessment, this approach provides a second method for pollen viability detection.

Another phenomenon existing in the field of plant phenomics is that the major research efforts are concentrated on model construction, focusing on refining and innovating models to ascertain the “optimal model”, while little attention is placed on the best practices for measurement or prediction (Matese et al., 2024). Thus, this intense concentration may theoretically prove a model feasible but may fail to perform its expected function in practical applications. In the existing use of deep learning models for pollen viability detection, Tan et al. (2022) developed a cotton pollen viability detection model based on YOLOv5. Although the model achieved ideal results on the given dataset, it lacked testing and application in different scenarios and was not equipped with corresponding hardware for model application and automation implementation (Xie et al., 2024).

In this work, we developed a low-cost yet highly efficient method for pollen viability detection. The main improvements are summarized as follows (Fig. 1):

(1) A novel Maize Pollen Viability Index (MPVI) was developed for rapid quantitative assessment of pollen viability across maize varieties, validated against conventional in vitro pollen tube germination tests.
(2) A YOLOv10-CBAM detection framework was proposed, integrating the Convolutional Block Attention Module (CBAM) to enhance feature representation and achieve robust performance in complex microscopic scenes.
(3) A Pollen Microscopic Image Segmentation Algorithm (PMISA) was proposed to removes complex backgrounds, and when combined with the Residual Dual-Attention Gated Network (RDAGNet), it overcomes the suboptimal performance of standalone YOLOv10-CBAM under high-density and low-quality image conditions.
(4) A low-cost PVDT device and AIpollen software were developed and validated on 350 maize inbred lines and 10 additional plant species, enabling cross-species pollen viability detection.
(5) Candidate genes for maize pollen heat tolerance were identified via Genome-Wide Association Study (GWAS), providing valuable genetic resources for heat-resilient maize breeding.

2. Materials and methods

2.1. Sample collection and pollen viability determination

The experiment was conducted in Zhuozhou, Hebei Province (115°58′E, 39°28′N) and Beijing (116°28′E, 40°02′N) during summer, and in Sanya, Hainan Province (109°30′E, 18°14′N) during winter. These sites cover distinct climates, soil types and cropping conditions, ensuring the diversity of the dataset and providing a solid foundation for validating the generalization ability of the model. For pollen collection, vigorous, pest- and disease-free plants with approximately one-third of their tassels shedding pollen were selected between 9 a.m. and 11 a.m.

In Zhuozhou, a total of 30 sampling plants were selected for each of 11 varieties (V1-V11, Supplementary Table 1). Pollens collected from 30 plants of each variety was immediately divided into five portions and maintained separately at 28 °C, 31 °C, 34 °C, 37 °C, and 40 °C. After 2 h, 4 h, 6 h and 8 h intervals, a 0.01 g fraction of pollen was extracted for viability determination. In Sanya, a population containing 350 inbred lines (Fu et al., 2013) was planted. Pollens from 10 plants per line was mixed, maintained at 28 °C and 40 °C, and a 0.1 g fraction was used for determination pollen viability.

The pollen vitality determination method based on TTC staining. The TTC staining method is a simple and rapid technique for determining viability, relying on the principle that living tissues with dehydrogenase activity reduce the colorless TTC to a red formazan dye (Yang et al., 2023). To validate its reliability for maize pollen, we compared it to the I2-KI staining and the in vitro pollen germination method using nine varieties of maize pollens (V1-V9, Supplementary Table 1). The specific protocol is detailed in Supplementary 1.1.

2.2. Collection and labeling of pollen microscopic images

Microscopic images of stained pollen were captured using a portable microscope system (3R-MSUSB390, Japan), with parameters: Field of view (FOV) = 2 mm, Magnification = 106.2 X, Working distance = 256 mm, Brightness = 10 (built-in light source), and Resolution set = 4 K. Images were saved in PNG format. Each sample was photographed three times under different view field. Based on the color and sharp of pollens, four types of viability were defined: high-viability pollen (HVP), red color and round shapes; low-viability pollen (LVP), pink color and round shapes; non-viability pollen (NVP), colorless and round shapes; and cracked pollen (CP), green color and irregular shapes.

The stained images were labeled using the Labelme software (MIT, USA). Pink, orange, red and yellow were used to represent HVP, LVP, NVP, and CP, respectively. Areas beyond the labels were considered as background. Subsequently, the labeling data was saved in JSON format, which was further converted into the YOLO dataset format. These labeled datasets (with 1551 pollen grains) were available at GitHub: https://github.com/HekeLi/pollen.

2.3. Proposal and validation of the MPVI

Traditional method only used the proportion of viable pollen grains to total grains for indicating viability, which fails to account for the quantity distribution and respective weights between HVP and LVP. HVP plays a more critical role in plant reproduction than LVP, so we introduced the principle of weighting and designed three MPVI (Formulas 1–3), for effectiveness testing. To validate the robustness and generalizability of the optimized MPVI, leave-one-variety-out cross-validation

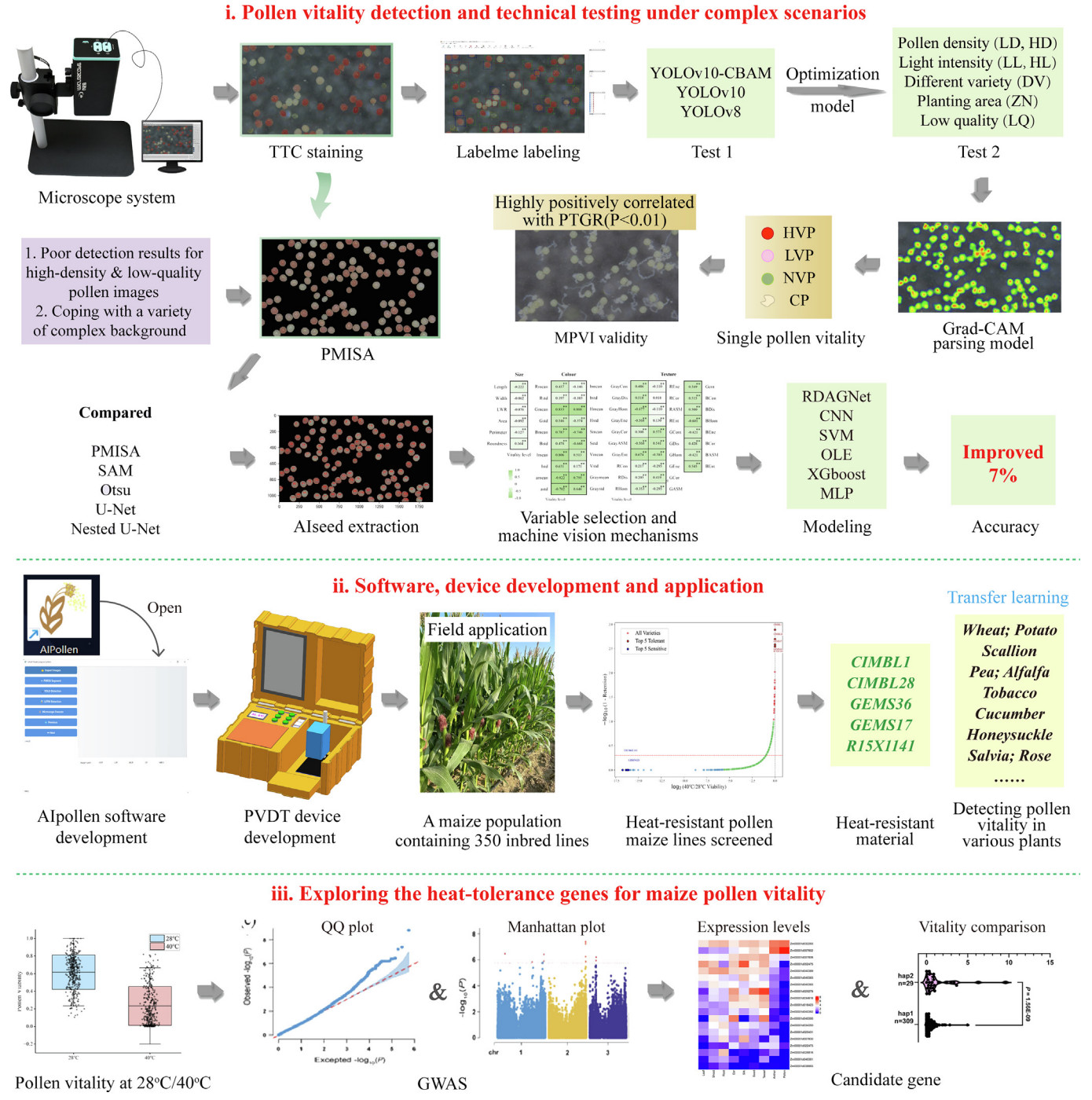


Fig. 1. The overall framework of this study.

(LOVOCV) was performed across 11 maize varieties based on Spearman's correlation analysis between multiple MPVI values and the in vitro Pollen Tube Germination Rate (PTGR, Formula 4). Sensitivity analysis and 95% confidence intervals (CI) were also calculated to evaluate the stability of parameter settings.

$$MPVI{-}1 = \frac{m * HVP + n * LVP}{\mathrm{HVP} + \mathrm{LVP} + \mathrm{NVP} + \mathrm{CP}} \quad (1)$$

$$MPVI{-}2 = \frac{HVP + \alpha * LVP}{\mathrm{HVP} + \mathrm{LVP} + \mathrm{NVP} + \mathrm{CP}} \quad (2)$$

$$MPVI{-}3 = \frac{\beta * HVP + LVP}{\mathrm{HVP} + \mathrm{LVP} + \mathrm{NVP} + \mathrm{CP}} \quad (3)$$

$$PTGR = \frac{M}{\mathrm{N}} \quad (4)$$

In Formula 1 to 3, the value of m ranges from 1 to 2 with a step size of 0.1, the value of n ranges from 0 to 1 with a step size of 0.1, subject to the condition that $m + n = 2$. The value range for α is 0 to 1 with a step size of 0.1, and the value range for β is 1 to 6 with a step size of 0.5. M represents the number of germinated pollen grains, and N represents the total number of pollen grains.

2.4. Method

We proposed two methods for detecting pollen viability to be applied in different scenarios. The first method involves direct detection using YOLOv10-CBAM. The second method entails extracting pollen with PMISA, followed by detection using RDAGNet classification model.

2.4.1. Pollen viability detection based YOLOv10-CBAM

2.4.1.1. Dataset preparation. Microscopic pollen images were collected from six maize varieties (V1–V6, Supplementary Table 1) with strict independent partitioning to ensure data reliability. All images from the same sampling batch were assigned to only one subset (training, validation, or test set) to avoid source similarity and performance overestimation. The training set and the first-level test set (Test1) were split at a ratio of 3:1, with 20% of the training set randomly selected as the validation set for model optimization. To further verify the model's generalization ability, a second-level external independent test set (Test2) was constructed, including different pollen densities, light conditions, image qualities, maize varieties, and planting areas: (1) Pollen density: high-density (Test2-HD, >50 pollen grains) versus low-density (Test2-LD, <50 pollen grains); (2) Light conditions: High-intensity (Test2-HL, level 12 brightness) and low-intensity (Test2-LL, level 8 brightness) relative to standard conditions (Level 10 brightness) (The level was determined by the nonlinear brightness control system of the microscopic imaging software (3R-MSUSB390, Japan) to simulate lighting variations; (3) Different varieties: (Test2-DV, V7-V11); (4) Image quality: Low-quality images (Test2-LQ, <500 KB) relative to standard conditions (normal-quality images >2000 KB), testing resilience against compression and distortion; and (5) Different planting areas: (Test2-ZN, the Science and Technology Park of China Agricultural University, Beijing (40°02′N, 116°28′E) relative to standard conditions (Zhuozhou, Hebei Province) to assess performance across different ecological settings.

2.4.1.2. Introduction to YOLOv10-CBAM. YOLOv10 is a real-time object detection system featuring the lightweight yet powerful EfficientRepV3 backbone network, which optimizes multi-level feature extraction (Wang et al., 2024). A key component is the Pyramid Split Attention (PSA) technology, which enriches dimensional information through refined multi-scale feature extraction and attention weight allocation (Zhang et al., 2022). However, PSA can overemphasize non-critical information when input data is noisy. To reduce the interference of noise and improve the feature extraction ability under complex pollen imaging scenarios, we proposed the YOLOv10-CBAM version (Fig. 2). CBAM was selected for its lightweight plug-and-play structure, which combines channel attention (CA) and spatial attention (SA) to adaptively focus on key pollen features while suppressing background noise (Woo et al., 2018). The CBAM module was inserted after the C2f modules in the backbone network of YOLOv10, which can enhance multi-scale feature representation without introducing excessive computational overhead. To verify the necessity and stability of CBAM, ablation experiments were conducted with four groups: (1) YOLOv10; (2) YOLOv10-CA; (3) YOLOv10-SA; (4) YOLOv10-CBAM. Additionally, we adopt the pre-trained YOLOv10-S model, which provide an ideal balance speed and accuracy. Training parameters for YOLOv10-CBAM were set as: Epochs to 500, Batch Size to 4, and image size to 1280. For comparative analysis, YOLOv8, YOLOv11, YOLOv12, Faster R-CNN, and EfficientDet were also evaluated. The detailed algorithms are described in Supplementary 1.2.

2.4.2. Pollen viability detection based PMISA-RDAGNet

2.4.2.1. Introduction to PMISA. The PMISA algorithm was proposed to segment TTC stained pollen images (Fig. 3a-3d). The algorithm workflow involves four main steps: (1) Circular Hough Transform: Detects pollen grains by analyzing edge information and accumulating votes in parameter space. A circle is identified when votes for a specific center-radius combination exceed a threshold corresponding to the minimum diameter (Cherabit et al., 2012). This step was also tested on elliptical (scallion) and irregular (honeysuckle) shapes (Supplementary Fig. 1). (2) Active Contour Model (Snake model): optimizes pollen grain contours by using external forces to guide the contour toward the true boundary while internal forces maintain smoothness. The contour is iteratively adjusted until the energy function is minimized (Ronfard, 1994). (3) Morphological processing: uses dilation and erosion operations to reduce image noise and smooth pollen boundaries (Raid et al., 2014). (4) Watershed algorithm: segments closely connected (touching) pollen grains by treating the gradients image as a topographic map, drawing watershed lines where adjacent basins meet to divide the image into distinct regions (Vachier and Meyer, 2005).

The PMISA package was incorporated into a functional module for straightforward utilization. The preprocessing hyperparameters and setting process used in the PMISA are listed in Supplementary Table 2 and Supplementary 1.3. Furthermore, to ensure fair comparison, unparameter-optimized PMISA was compared with unsupervised segmentation algorithms (Segment Anything Model (SAM) and Otsu's thresholding method) under uniform fully automatic and prompt-free conditions. Meanwhile, the parameter-optimized PMISA is benchmarked against customized segmentation models (U-shaped convolutional neural network (U-Net) and Nested U-Net) developed on annotated pollen datasets as comparative benchmarks. SAM is a visual foundation model with powerful zero-shot generalization ability for arbitrary object segmentation, yet it highly relies on manual visual prompts (Kirillov et al., 2023). Otsu is a classic unsupervised thresholding algorithm that can automatically determine the optimal separation threshold between foreground and background (Goh et al., 2018). U-Net adopts a symmetric encoder-decoder structure and fuses shallow texture features with deep semantic information via skip connections to accomplish image segmentation tasks efficiently (Ronneberger et al., 2015). Nested U-Net adds nested skip connections on the basis of U-Net to strengthen multi-level feature interaction, which significantly improves the detail retention capability in complex structure segmentation (Zhou et al., 2018).

2.4.2.2. Establishment of pollen viability classification model. The segmented pollen images were extracted to machine vision indices, including the size, color, and texture of the pollen (54 in total, Supplementary Table 3). Spearman correlation analysis (SPSS23, IBM USA) identified the most significant index related to pollen viability. On this basis, we propose a Residual Dual-Attention Gated Network (RDAGNet) for pollen vitality classification (Supplementary Fig. 8). The network adopts three one-dimensional residual blocks to mine deep internal correlations of static features layer by layer, and relieves the gradient vanishing issue via residual shortcut connections. The last residual block introduces downsampling with a stride of 2 to compress feature dimensions and remove redundant information. A dual attention mechanism is embedded to adaptively weight key discriminative features from channel and spatial dimensions. Finally, the Gated MLP structure is adopted to replace the traditional fully connected layer, which improves feature screening accuracy and model generalization capability. To validate the superiority of RDAGNet, the proposed model was compared with classical classification models including Convolutional Neural Network (CNN), Multilayer Perceptron (MLP), Extreme Gradient Boosting (XGBoost), support vector machine (SVM), and optimized ensemble learning (OEL). In addition, ablation experiments were designed to verify the effectiveness of each module. The data was divided into training and test sets in a 3:1 ratio using the Kennard-Stone algorithm, which employs Euclidean distance to extract the most representative sample points from the dataset (Jin et al., 2024). The

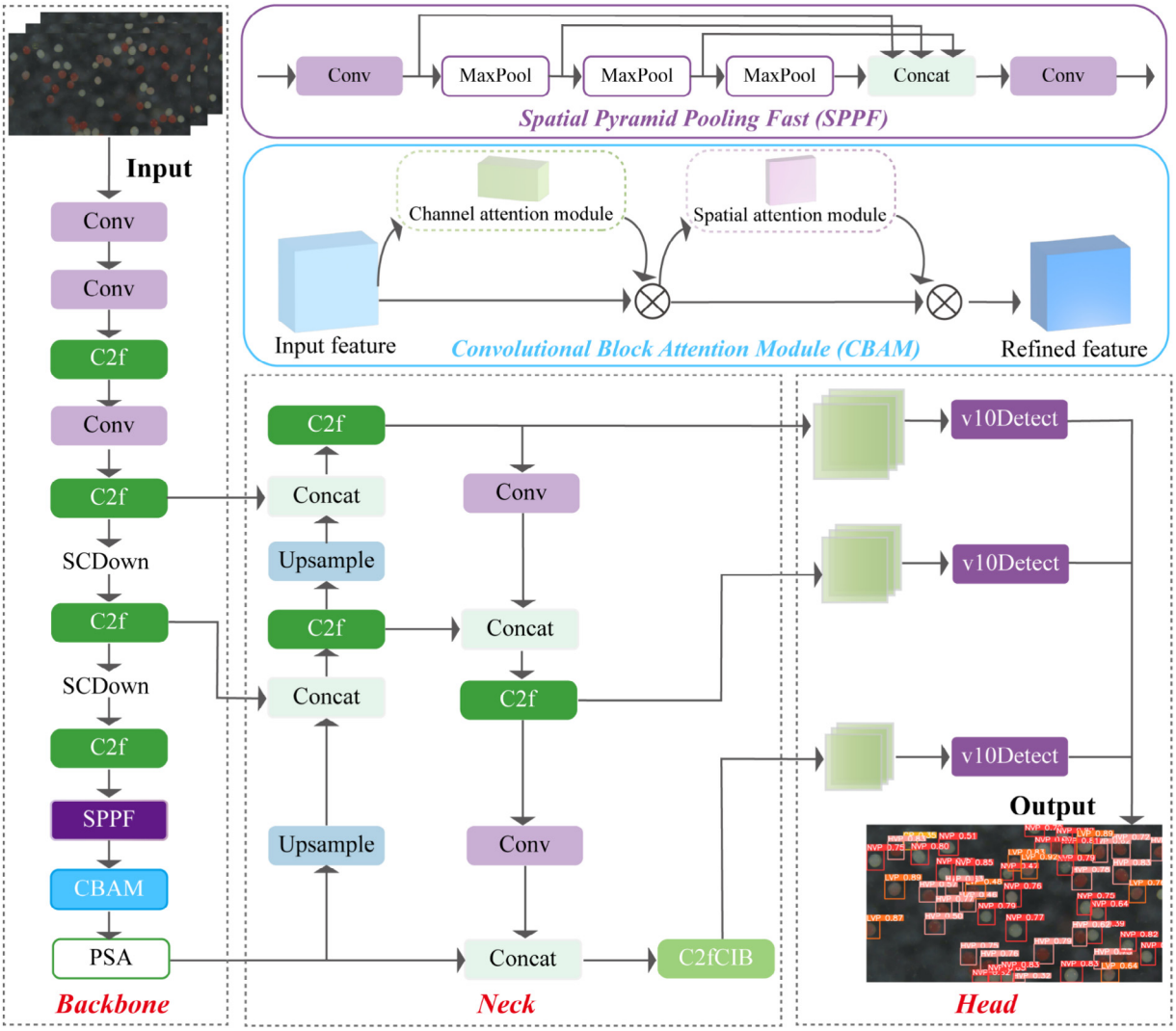


Fig. 2. Network structure of YOLOv10-CBAM. The network comprises three main modules: The backbone, which extracts multi-level features from the input image to provide foundational data; the Neck, which aggregates features from different backbone stages to generate multi-scale feature maps, enhancing detection of objects at varying scales; and the Head, which generates category and location predictions based on the Neck's output.

specific principles and training protocols for these algorithms are described in Supplementary 1.4.

2.5. Experimental environment and model evaluation

The hardware environment for processing data included an AMD Ryzen 76800H processor with Radeon Graphics at 3.20 GHz, and 16 GB of RAM, running on Windows 11. The software environment includes Python 3.9, PyTorch 2.0.1, and MATLAB 2023b.

Model performance in pollen viability detection was evaluated using several metrics: (1) Intersection over Union (IoU): used to define the matching between predicted bounding boxes and ground-truth boxes. A detection is regarded as valid when IoU ≥ 0.5; (2) AP_{50} (AP@0.5): average precision for each pollen category at an IoU threshold of 0.5; (3) mAP@0.5: mean average precision across all four pollen categories; (4) Accuracy: measures overall classification correctness; (5) Precision and Recall: assess the identification performance for different viability types; (6) F1-score: evaluates the comprehensive performance for each specific viability category. The formulas for the evaluation metrics are as follows:

$$IoU = \frac{A \cap B}{A \cup B} \tag{6}$$

$$AP_{50} = \int_0^1 p(r)dr \tag{7}$$

$$mAP@0.5 = \frac{\sum_{i=1}^{C} AP_{50}^{i}}{C} \tag{8}$$

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{9}$$

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

$$F1\text{—}score = 2 * \frac{precision * recall}{precision + recall} \tag{12}$$

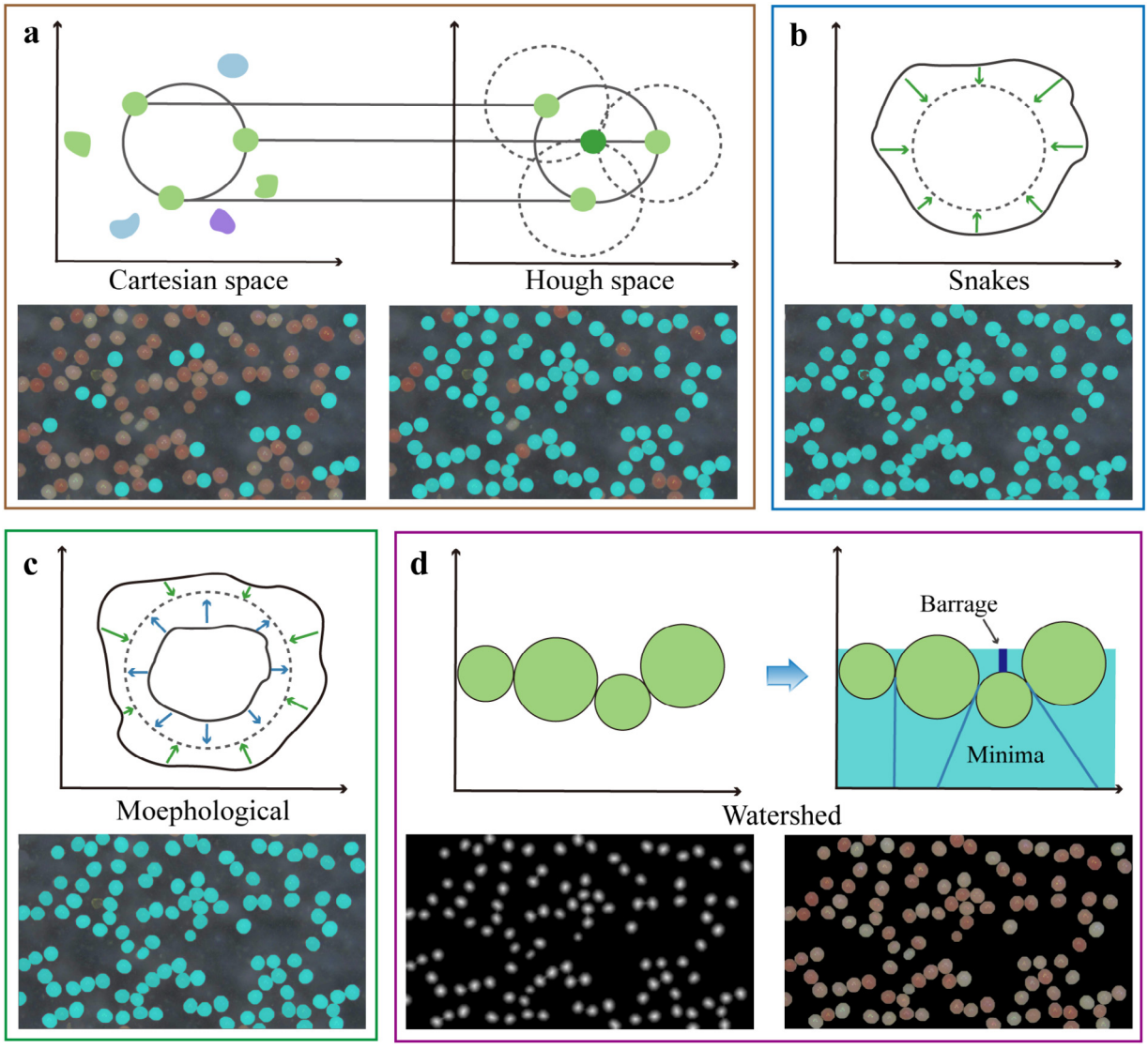


Fig. 3. PMISA algorithm segmentation workflow. (a) Application of the Circular Hough Transform to detect all pollen grains. The transition from Cartesian to Hough space is illustrated by blue circles, representing the pollen detection (background extraction) process. Subsequent processing targets these detected grains; (b) Snake model for optimizing pollen grain contours based on the initially detected blue circled pollen grains; (c) Morphological processing to remove image noise and smooth pollen boundaries; (d) Watershed algorithm for segmenting adhering pollen grains. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

here, A is the predicted bounding boxes or segmentation mask, and B is the ground-truth boxes or segmentation mask; $p(r)$ represents precision as a function of recall; C is the total number of pollen viability categories, and i denotes the i-th category; TP (true positive) is the number of correctly identified positive cases; TN (true negative) is the number of correctly identified negative cases; FP (false positive) is the number of incorrectly identified positive cases; FN (false negative) is the number of incorrectly identified negative cases (Cardellicchio et al., 2025; Li et al., 2025a, 2025b).

2.6. Development of the AIpollen software and PVDT device

To provide breeders with a portable and efficient auxiliary tool for fieldwork, we have developed a pollen viability detection tool (PVDT). PVDT integrates the intelligent detection software (AIpollen), a temperature control system, and an image acquisition system. The AIpollen software, which includes two distinct pollen viability detection models (YOLOv10-CBAM and PMISA- RDAGNet) was written in Python 3.9.5. The GUI was developed using PySide6 6.4.0. Backend image processing relies on OpenCV 4.5.5 and Scikit-image 0.19.2. while visualization and plotting were conducted using Matplotlib 3.5.1 and Seaborn 0.12.2.

3. Results

3.1. Analysis and comparison of MPVI

To validate the reliability of TTC staining method for maize pollen, we compared it to the I_2-KI staining. Under optimal conditions (28 °C), TTC results (Stained pollen count/Total pollen count) correlated significantly with I_2-KI ($R = 0.83$, $p < 0.01$) (Supplementary Fig. 2a-2c). However, under high temperature (40 °C), TTC accurately indicated complete loss of viability (no staining), whereas I_2-KI falsely suggested retained activity (dark staining due to intact starch; Supplementary Fig. 2d,2e). This suggests that dehydrogenase activity (metabolic function) detect method TTC is superior than the maturity (starch accumulation) detection method I_2-KI for referring pollen viability. Then, we compared the TTC results to the in vitro Pollen Tube Germination Rate (PTGR), which showed significantly correlations between two results

at both 28 °C ($R = 0.74$, $p < 0.01$) and 40 °C ($R = 0.85$, $p < 0.01$) (Supplementary Fig. 2f-2 k). However, the in vitro germination test required at least 2 h, while the TTC test only required 0.5 h. Together, these results suggested that TTC staining is an accurate and efficient pollen viability test method.

To enhance the predictive accuracy of TTC, three Maize Pollen Viability Indices (MPVI) were constructed based on the weighting principle. Leave-one-variety-out cross-validation indicated that MPVI-1 exhibited the highest correlation with the PTGR. Specifically, when parameters m and n were set to 1.5 and 0.5, respectively, MPVI-1 achieved the optimal correlation ($R = 0.820$, $p < 0.01$), with a 95% confidence interval of 0.769–0.860, demonstrating high reliability of this weighted index (Supplementary Fig. 3d). Sensitivity analysis showed that slight variations in m and n did not cause significant changes in correlation strength: the correlation coefficient was 0.817 at $m = 1.6$, $n = 0.4$ and 0.816 at $m = 1.4$, $n = 0.6$ (Supplementary Fig. 3a–3c). These results confirm that the MPVI index is robust and generalizable for evaluating pollen viability across diverse maize materials. Similarly, MPVI-2 and MPVI-3 performed best when HVP weight ($\beta = 3.0$) significantly exceeded LVP weight ($\alpha = 0.3$), confirming that with appropriate parameters (HVP contributes three times more to viability than LVP), MPVI can accurately indicate pollen viability. Then, we used the MPVI-1 ($m = 1.5$, $n = 0.5$) for 11 maize varieties (Supplementary Table 1) to evaluate the effect of temperature on pollen viability. Results showed that pollen viability gradually decreased with increasing temperature and processing time (Supplementary Fig. 3e-3i). Varieties V1, V4, and V5 exhibited a relatively slow decrease, demonstrating superior heat tolerance. Notably, pollen viability peaked at 28 °C and 31 °C for all tested varieties, indicating optimal temperatures for keeping maize pollen viability.

3.2. Development of YOLOv10-CBAM for evaluating MPVI

Although MPVI can accurately predict pollen viability under various treatment temperature and process time, it is operationally complex and prone to subjective error during manual classification. To address this, we developed YOLOv10-CBAM to rapidly detect pollen vitality and conducted tests on Test1 (pollen dataset of the same variety) and Test2 (pollen dataset of different varieties, densities, brightness, planting areas, and low quality). To systematically evaluate the performance of YOLOv10-CBAM, we compared it with YOLOv10, YOLOv8, YOLOv11, YOLOv12, Faster R-CNN, and EfficientDet models.

3.2.1. Detection results of the model on the Test1 dataset

The detection results of different models on the Test1 dataset showed that YOLO series models outperform Faster R-CNN and EfficientDet (Table 1, Fig. 4d-2f, Supplementary Fig. 4). Specifically, the mAP@0.5 values of YOLOv10-CBAM, YOLOv10 and YOLOv11 reach 97.5%, 96.1% and 95.2%, respectively, demonstrating that all models achieve satisfactory detection and classification precision. Among, YOLOv10-CBAM obtains the highest classification accuracy of 96.9%, followed by YOLOv10 with 95.9%. Regarding specific pollen types, YOLOv10-CBAM demonstrated superior performance for HVP, LVP, and CP (F1-scores: 97.1%, 94.5%, and 99.2%, respectively). Notably, CP obtains the best overall recognition results, with a precision of 100% and a recall of 98.4%. By comparison, YOLOv10 performed best in recognizing NVP (F1-score: 98.8%). To explore the contribution of each attention module in CBAM to the YOLOv10 model, ablation experimental results showed that YOLOv10-SA and YOLOv10-CA achieved mAP@0.5 values of 96.7% and 96.2%, respectively, both outperforming the baseline YOLOv10 (96.1%) (Supplementary Table 4). This verifies that both spatial attention and channel attention modules can independently improve the detection performance of the model. For the CP category, YOLOv10-CBAM reached an F1-score of 99.2%, which is obviously higher than YOLOv10-SA (95.8%) and YOLOv10-CA (97.2%). It demonstrates that the fusion of spatial and channel attention can significantly enhance the classification accuracy of CP pollen.

To assess model interpretability, we generated feature heatmaps using Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017). According magnified local image regions, visual analysis revealed that both YOLOv10 and YOLOv10-CBAM captured subtle features, such as faint pollen grain textures at image edges that could not be detected by YOLOv8 (Fig. 4g-4i). Furthermore, the YOLOv10-CBAM heatmaps displayed denser hotspot distributions and deeper red regions than other models. These results confirm that the CBAM attention mechanism effectively enhances the model's focus on

Table 1
Detection results of different models on the Test1 dataset.

Model	Category	Accuracy (%)	mAP@0.5 (%)	AP_{50} (%)	Precision (%)	Recall (%)	F1-Score (%)
YOLOv10-CBAM	HVP	96.9	97.5	97.3	97.5	96.8	97.1
	LVP			94.9	94.5	94.5	94.5
	NVP			98.8	98.7	98.4	98.5
	CP			99.1	100	98.4	99.2
YOLOv10	HVP	95.9	96.1	95.8	96.4	95.2	95.8
	LVP			91.6	91.8	93.1	92.4
	NVP			98.5	99.1	98.5	98.8
	CP			98.6	100	96.7	98.3
YOLOv8	HVP	89.4	92.3	93.1	95.5	88.8	92.0
	LVP			84.6	84.8	85.3	85.1
	NVP			95.7	96.6	92.5	94.5
	CP			95.8	100	90.2	94.8
YOLOv11	HVP	95.1	95.2	96.3	97.4	96.2	96.8
	LVP			92.9	93.3	92.8	93
	NVP			96.8	98.1	96.3	97.2
	CP			94.8	100	91.8	95.7
YOLOv12	HVP	91.7	92.6	94.3	97.6	94.1	95.8
	LVP			88.2	92.6	87.2	89.8
	NVP			94.1	97.7	92.9	95.2
	CP			93.6	98.2	90.2	94
Faster R-CNN	HVP	88.9	91.9	93.1	95.9	88.5	92.6
	LVP			85.3	85.9	85.9	85.9
	NVP			94.8	98.2	90.5	94.2
	CP			94.2	100	88.5	93.9
EfficientDet	HVP	88.2	90.9	90.8	95.4	87.8	91.4
	LVP			85.7	90.5	84.5	87.4
	NVP			93.3	97.8	90.7	94.1
	CP			93.9	100	90.2	94.8

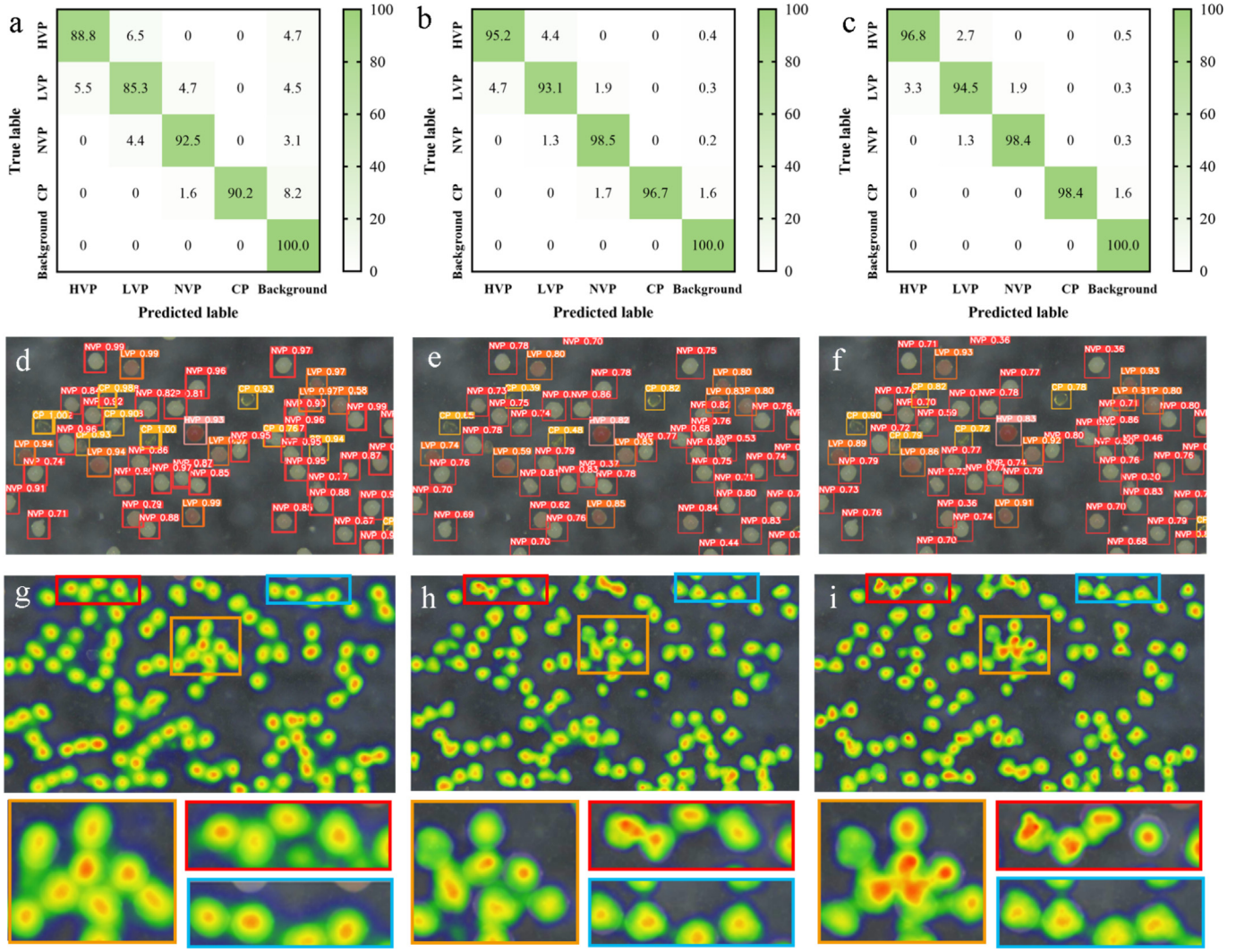


Fig. 4. The detection results of YOLOv10-CBAM, YOLOv10, and YOLOv8 models on the Test1 dataset. (a-c) Confusion matrices for YOLOv8 (a), YOLOv10 (b), and YOLOv10-CBAM (c) on the Test1 dataset; (d-f) Detection visualization results for YOLOv8 (d), YOLOv10 (e), and YOLOv10-CBAM (f); (g-i) Grad-CAM feature heatmaps for YOLOv8 (g), YOLOv10 (h), and YOLOv10-CBAM (i). Note: The color gradient from light green to deep red visually represents the contribution of specific image regions to the model's prediction. Deeper red hues indicate higher feature weight and greater significant in the model's decision-making process. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

critical features, improving prediction reliability and robustness. Given their outstanding performance on Test1, both YOLOv10, YOLOv11, and YOLOv10-CBAM were selected for further evaluation on Test2 dataset.

3.2.2. Detection results of the model on the Test2 dataset

To evaluate the model performance for detecting pollen viability across different scene images, we firstly focused on the Test2-LD (Low density: <50 grains) dataset. The results showed that among the three models (YOLOv10-CBAM, YOLOv10, and YOLOv11), YOLOv10-CBAM exhibited the best performance, achieving an mAP@0.5 of 99.2% and an accuracy of 98.5%. This outperformed YOLOv11 (mAP@0.5: 97.5%, accuracy: 97.0%) and YOLOv10 (mAP@0.5: 98.6%, accuracy: 97.7%) (Fig. 5a). Additionally, all three models demonstrated more effective classification performance for LVP, NVP, and CP than for HVP (Supplementary Table 5). On the Test2-HD (High density: >50 grains) dataset, the mAP@0.5 and accuracy of all three models decreased significantly by 3–5% compared to those on Test2-LD. Specifically, YOLOv10-CBAM achieved the highest mAP@0.5 of 92.6%, followed by YOLOv11 (92.3%) and YOLOv10 (91.8%). In terms of accuracy, YOLOv10-CBAM still maintained the highest value of 91.5%, followed by YOLOv11 (91.4%) and YOLOv10 (90.8%). The confusion matrix indicated that misclassification primarily occurred between adjacent categories, particularly between HVP and LVP (Fig. 5c).

The test results on the Test2-LL (Low light: level 8 brightness; standard conditions: level 10 brightness) dataset showed that YOLOv10-CBAM achieved the best performance with an mAP@0.5 of 96.2%, outperforming YOLOv10 (93.3%) and YOLOv11 (91.5%) (Fig. 5a, Supplementary Table 6). In terms of classification accuracy, YOLOv10 and YOLOv11 dropped by 4% and 5%, respectively, compared with the Test1 baseline, while YOLOv10-CBAM maintained a relatively stable accuracy of 94.2%. On the Test2-HL (High light: level 12 brightness) dataset, all models presented a 2–5% decline in mAP@0.5 and accuracy compared with the Test1 dataset. Specifically, YOLOv10-CBAM still maintained superior comprehensive performance, yielding an mAP@0.5 of 95.8% and an accuracy of 93.9%. By comparison, YOLOv10 achieved an mAP@0.5 of 93.0% and an accuracy of 91.0%, and YOLOv11 obtained an mAP@0.5 of 92.3% and an accuracy of 90.7%. The confusion matrix showed that the majority of misclassifications occurred between adjacent pollen categories, particularly between HVP and LVP, and between LVP and NVP (Fig. 5e).

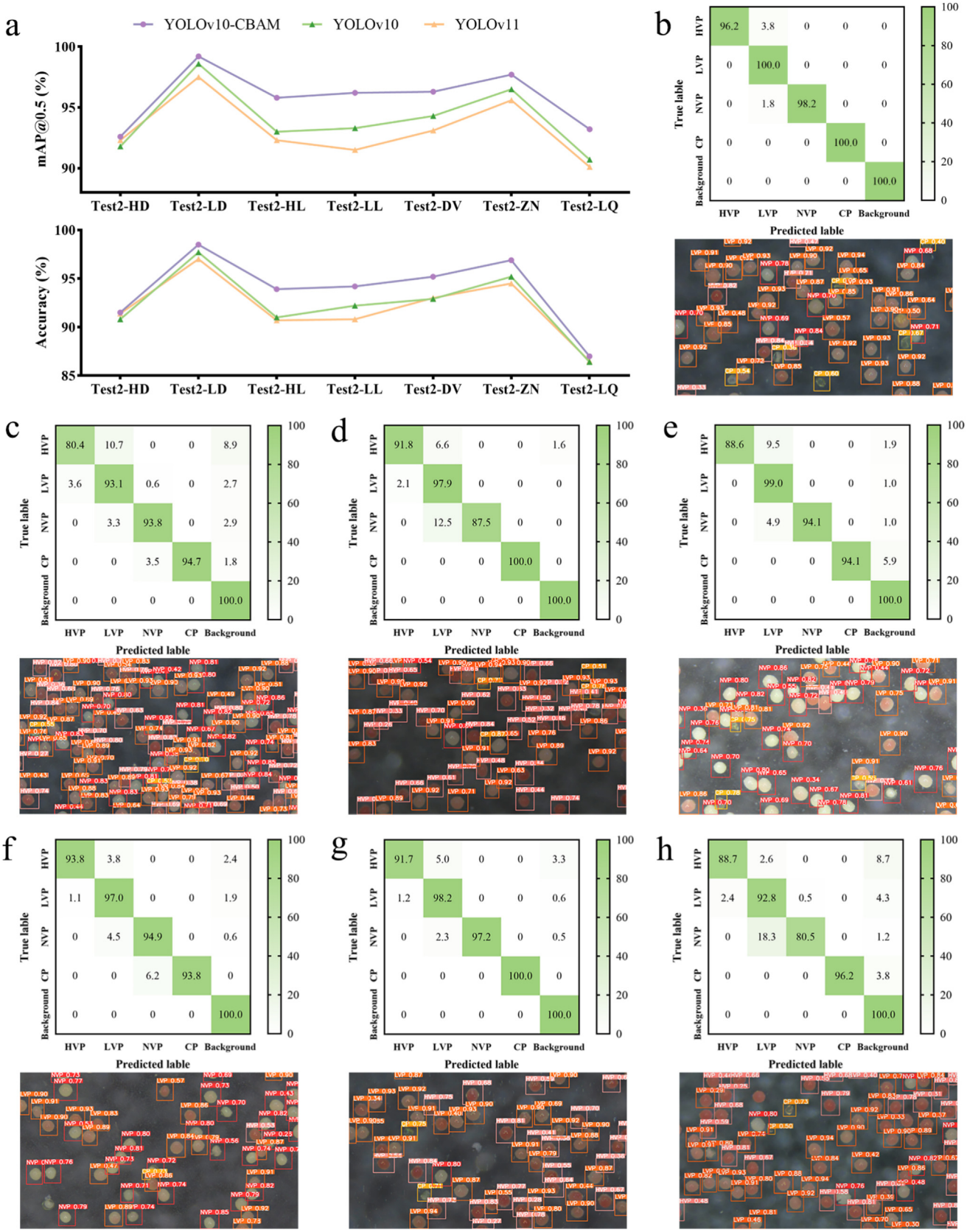


Fig. 5. Test results of the model on the Test2 dataset. (a) Accuracy and mAP@0.5 of the YOLOv10-CBAM, YOLOv10, and YOLOv11 model; (b-h) Confusion matrices and recognition images of YOLOv10-CBAM detection results in different subsets of Test2, which are Test2-LD (b), Test2-HD (c), Test2-LL (d), Test2-HL (e), Test2-DV (f), Test2-ZN (g), and Test2-LQ (h) in sequence.

The test results on the Test2-DV (Different varieties: V7-V11) dataset showed that YOLOv10-CBAM achieved an mAP@0.5 of 96.3% and an accuracy of 95.2%, surpassing YOLOv10 (mAP@0.5: 94.3%, accuracy: 92.9%) and YOLOv11 (mAP@0.5: 93.1%, accuracy: 93.0%) (Fig. 5a). Among the three models, YOLOv10-CBAM also demonstrated better classification performance for all pollen types, particularly for NVP and CP (F1 scores: 97.2% and 96.8%, respectively), compared to YOLOv10 (F1 scores: 95.7% and 93.3%, respectively) and YOLOv11 (F1 scores: 95.7% and 90.3%, respectively) (Supplementary Table 7). To validate cross-variety applicability, we analyzed pollen morphology (Area, Roundness, and length-width ratio (LWR)) across 11 varieties (Supplementary Fig. 5). One-way analysis of variance (ANOVA) and Duncan's MRT (multiple range test) revealed no significant differences ($p > 0.01$), indicating high morphological consistency. These results suggest that the model's detection performance is robust against genotypic differences.

The test results on the Test2-ZN (Different planting areas: Beijing; standard conditions: Hebei) dataset showed that YOLOv10-CBAM achieved stable performance across the two planting areas with an mAP@0.5 of 97.7% and an accuracy of 96.9% (Fig. 5a), particularly for the classification of CP (F1 score 100%) (Supplementary Table 8). It outperformed YOLOv10 (mAP@0.5: 96.5%, accuracy: 95.2%) and YOLOv11 (mAP@0.5: 95.6%, accuracy: 94.5%). However, on the Test2-LQ (Low-quality image: <500 KB resolution; standard conditions: >2000 KB) dataset, all models showed a 4–7% performance decline compared with the Test1 dataset. Specifically, YOLOv10-CBAM obtained the best performance with an mAP@0.5 of 93.2% and an accuracy of 87.0%, followed by YOLOv10 (mAP@0.5: 90.7%, accuracy: 86.4%) and YOLOv11 (mAP@0.5: 90.1%, accuracy: 86.5%). The confusion matrix indicated that classification errors were mainly concentrated among adjacent pollen categories, especially between HVP and LVP, and between LVP and NVP (Fig. 5h).

3.3. Development of PMISA-RDAGNet to improve pollen viability detection accuracy in high-density pollen and low-quality images

To overcome the unsatisfactory of YOLOv10-CBAM model in handing high-density pollen (accuracy: 91.5%) and low-quality images (accuracy: 89.4%) (Fig. 5a), we developed the PMISA to eliminate complex backgrounds and effectively segment adhered pollen clusters in TTC-stained images.

3.3.1. Performance evaluation of PMISA segmentation

To systematically evaluate the performance of PMISA, we compared it with SAM, Otsu, U-Net, and Nested U-Net. The results show that the unparameter-optimized PMISA achieves an average accuracy of 91.9% and an IoU of 91.2%, slightly outperforming SAM (accuracy: 91.1%; IoU: 90.6%) and significantly outperforming Otsu (accuracy: 72.7%; IoU: 70.4%) (Supplementary Table 9). The parameter-optimized PMISA achieves a segmentation accuracy of 98.7% and a recall of 99.8%, demonstrating high accuracy and reliability in extracting pollen grains (Fig. 6a-6b). Its accuracy significantly exceeds that of U-Net (81.6%) and Nested U-Net (82.7%). This low accuracy was primarily due to their inability to separate overlapping pollen grains, which rendered them unsuitable for high-density pollen segmentation tasks.

3.3.2. The classification results of the pollen viability model

Following segmentation, we extracted 54 machine vision indices related to pollen size, color, and texture. Spearman correlation analysis was used for feature selection (variable selection: $R>0.5$ and $p<0.01$). Based on the selected variables, we established pollen viability classification models using RDAGNet, MLP, XGBoost, CNN, SVM, and OEL. Correlation analysis showed that most morphological features exhibited weak to moderate correlations with pollen viability. The length feature achieved the relatively strongest correlation with viability level (−0.222; Fig.6c), though the overall correlation magnitude was low. For color indicators, Gmean showed the highest correlation (0.833; Fig. 6c), far surpassing other color indicators. For texture indicators, the maximum correlation coefficient to pollen viability was GrayHom (0.677; Fig. 6c). We also analyzed the probability density distribution plots for length, Gmean, and GrayHom, which confirmed that no single indicator was sufficient to accurately distinguish viability level of pollen (Fig. 6f-6h).

Classification models were tested on the challenging Test2-HD dataset and Test2-LQ datasets. In both datasets, the RDAGNet model exhibited the best overall performance. On the Test2-HD dataset, RDAGNet achieved the best accuracy rate at 97.9% (Table 2, Fig. 6d), while the CNN model followed but only had an accuracy of 87.9%. On the Test2-LQ dataset, RDAGNet also achieved the best accuracy at 97.1% (Fig. 6e), when the CNN model achieved the second accuracy at 88.8%. The analysis of accuracy, recall, and F1-score indicated that on the Test2-HD dataset, RDAGNet achieved the highest F1-scores for HVP, LVP and NVP classification, with levels of 98.6%, 96.9%, and 98.4%, respectively. In the Test2-LQ dataset, RDAGNet maintained the highest F1-scores for HVP, LVP, and NVP classification, with levels of 98.6%, 90.6% and 96.9%, respectively, while LVP had the worst classification performance. In summary, deep learning models (RDAGNet and CNN) generally outperformed traditional machine learning models in pollen detection. The ablation experiment results demonstrate that, on the Test2-HD dataset, ResNet_CA and ResNet_SA achieve accuracies of 95.3% and 92.6% respectively, both outperforming the baseline model ResNet (90.1%, Supplementary Table10). On the Test2-LQ dataset, ResNet_CA and ResNet_SA achieve accuracies of 93.3% and 93.6% respectively, also both surpassing the baseline model ResNet (92.2%). This proves that both the spatial attention module and the positional attention module can individually enhance model detection performance. For the HVP and LVP categories, RDAGNet achieves F1 scores of 98.6% and 96.9% respectively, significantly higher than ResNet_CA (HVP: 94.2%, LVP: 93.4%) and ResNet_SA (HVP: 90.5%, LVP: 89.4%). This indicates that the fusion of spatial attention and positional attention can improve classification accuracy for HVP and LVP categories.

To verify whether the PMISA algorithm would generate false-positive results during the segmentation of CP, we conducted a separate test using 20 high-density images containing >15 CP pollens. The results indicated that PMISA achieved a precision of 99.1% and a recall of 99.7% for CP processing tasks (Supplementary Fig. 6a-6d). In addition, it attained 99.9% precision and 99.8% recall for non-CP pollens. These results demonstrates that the vast majority of CP samples can be effectively screened out, implying that the improved accuracy of the RDAGNet model is not due to the exclusion of CP samples, but rather to the enhanced ability to accurately classify non-CP samples.

3.4. Development of AIpollen software and PVDT device

To facilitate the use by researchers, we developed a detecting pollen viability software (AIpollen) (Fig. 7c). The graphical user interface (GUI) of this software consists two main components: (1) image acquisition or loading, and (2) viability analysis (Fig. 7a). In detail, after launching the software (Step 1), users have two options (Step 2): either driving the microscope to scan images or directly loading images (selecting "Load Folder" will analyze all images in the chosen folder). Once pollen images are acquired or opened, Step 3 allows users to select measurements from "YOLO Detection," "PMISA Segmentation," or "RDAGNet Detection" based on specific analysis tasks. Step 4 outlines two strategies for pollen viability detection. The first strategy uses the YOLO detection module to directly assess pollen viability and calculate the MPVI index based on the count of pollen categories. The second strategy employs the PMISA segmentation module to segment pollen images, extract color metrics for individual pollen grains, and then use RDAGNet detection to compute the viability of individual pollen and the MPVI index.

To provide breeders with a portable and efficient auxiliary tool for fieldwork, we developed a PVDT that composes the AIpollen software,

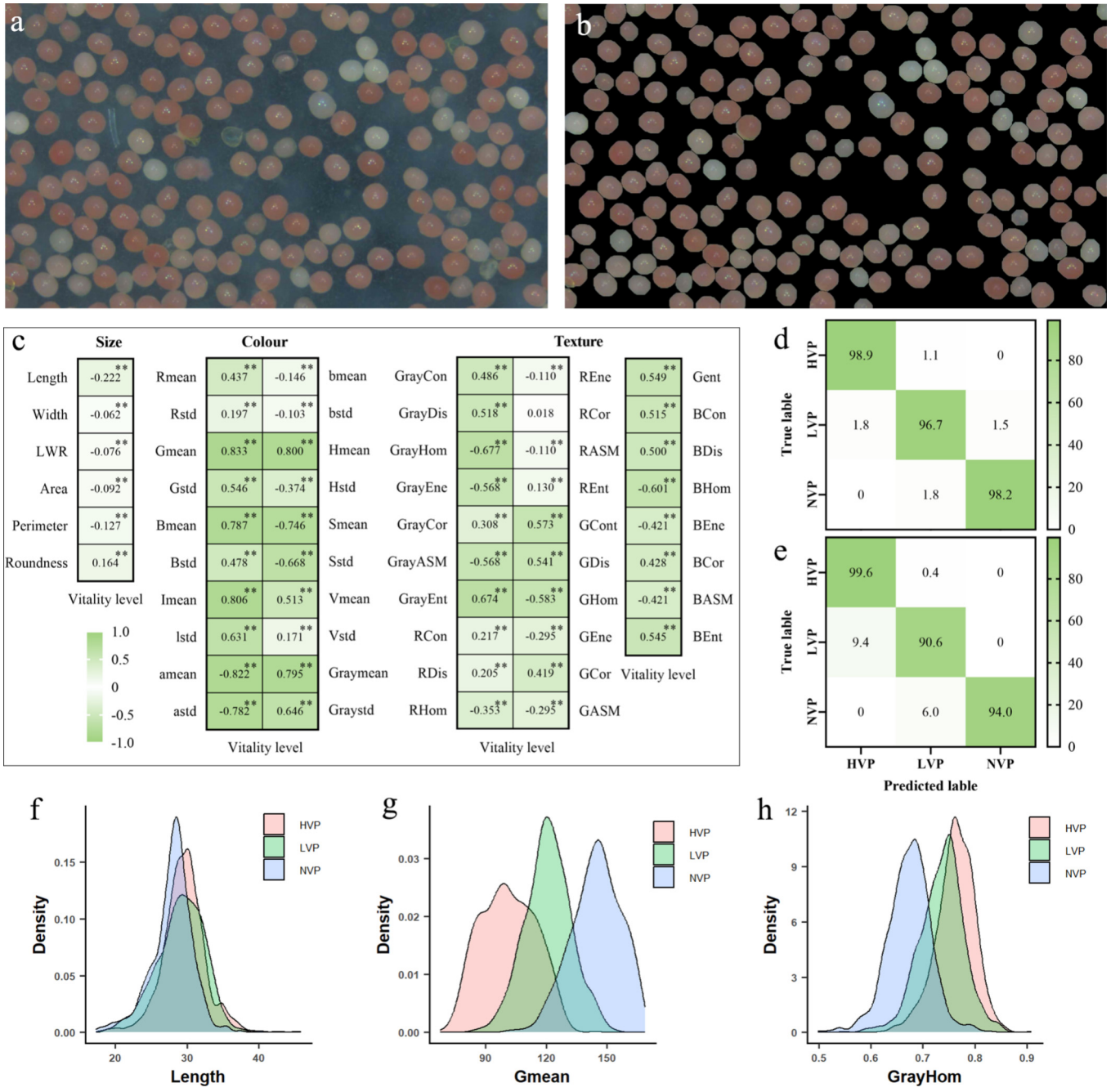


Fig. 6. The PMISA- RDAGNet based pollen viability detection. (a-b) Original pollen image at high density and the image after segmentation using PMISA; (e) Correlation analysis between size, color, and texture indices and pollen viability. ** indicated a highly significant relationship between machine vision metrics and vigor level ($p < 0.01$); (f, g) Confusion matrixes for the optimal RDAGNet model on the Test2-HD and Test2-LQ datasets. (h-j) Probability density distribution plots for length, Gmean, and GrayHom indicators.

a microscopic imaging system, an intelligent temperature control unit, biochemical reaction components, and auxiliary devices (Fig. 7b). The device adopts a modular design, with all components housed in a portable case meeting IP67 protection standards, ensuring reliability and ease of operation in field conditions. And it operates using a standardized procedure:

Sampling: Collect in-situ pollen samples using a handheld pollen sampler.

Staining: Transfer the collected samples into a dedicated reagent kit pre-filled with the staining reaction solution.

Incubation: Seal the sample kit and place it in the intelligent temperature control unit for a 20-min enzymatic colorimetric reaction under a constant temperature of 28–31 ± 0.5 °C.

Slide Preparation: After temperature control, use a micropipette to aspirate 10 μL of the reaction solution and deposit it onto a slide for solid-phase preparation.

Imaging and Analysis: Capture microscopic images using an integrated digital microscope, then analyze both morphological and staining features of pollen via the AIpollen software for viability assessment.

3.5. Field application of AIpollen software and PVDT device

3.5.1. Field application of PVDT in exploring the heat-tolerance genes of maize pollen viability

To validate the applicability of the PVDT device, we assessed pollen viability in a maize population of 350 inbred lines (Fu et al., 2013)

Table 2
Model's classification results of pollen viability based on machine vision data.

Data type	Model	Category	Training set	Test set			
			Accuracy (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Test2-HD	CNN	HVP	90.4	87.9	88.4	88.4	88.4
		LVP			80.6	83.3	81.9
		NVP			93.6	90.6	92.1
	RDAGNet	HVP	98.6	97.9	98.2	98.9	98.6
		LVP			97.1	96.7	96.9
		NVP			98.5	98.2	98.4
	SVM	HVP	85.9	83.0	84.1	78.6	81.2
		LVP			74.2	77.5	75.8
		NVP			91.1	93.1	92.1
	OEL	HVP	86.6	82.7	83.2	81.1	82.1
		LVP			74.6	74.9	74.8
		NVP			90.0	92.0	91.0
	XGBoost	HVP	87.3	84.4	86.4	76.4	81.1
		LVP			77.2	80.0	78.6
		NVP			86.6	96.7	93.0
	MLP	HVP	89.4	85.8	86.7	75.6	80.8
		LVP			78.2	83.6	80.8
		NVP			92.8	98.2	95.4
Test2-LQ	CNN	HVP	90.3	88.8	99.0	94.1	96.5
		LVP			58.6	96.1	72.8
		NVP			100	66.5	79.9
	RDAGNet	HVP	98.5	97.1	97.8	99.6	98.6
		LVP			90.6	90.6	90.6
		NVP			100	94.0	96.9
	SVM	HVP	89.1	88.3	90.2	100	94.8
		LVP			76.7	35.9	48.9
		NVP			86.0	91.6	88.7
	OLE	HVP	87.2	84.7	88.0	89.6	88.8
		LVP			70.0	43.2	56.0
		NVP			59.4	100	74.6
	XGBoost	HVP	88.3	86.9	91.1	93.0	92.0
		LVP			60.7	57.8	59.2
		NVP			92.6	89.8	91.2
	MLP	HVP	90.2	88.7	92.7	93.5	93.1
		LVP			64.2	67.2	65.6
		NVP			96.2	89.8	92.9

under two temperature conditions: 28 °C (favorable) and 40 °C (high-temperature stress). Results showed that high temperatures significantly reduced pollen viability, from an average of 61.9% at 28 °C to an average of 23.3% at 40 °C (Fig. 8a). Analysis of the relationship between the extent of viability decline and retention rates revealed three distinct response patterns within the population (Fig. 8b). The largest group was highly sensitive, with pollen viability almost completely lost under high temperatures, showing decline values between 0.4 and 0.6 and retention rates near 0. The moderately sensitive group contained 45 lines, exhibiting a viability decline of approximately 0.2 and a retention rate around 0.8. Only 25 lines were classified as heat-tolerant, with viability decline close to 0 and retention rates near 1.

To optimize data normality and enhance statistical analysis power, we applied $\log_2$ transformation to the pollen viability change data between the two temperatures, allowing intuitive visualization of the direction and magnitude of temperature effects. Additionally, $\log_{10}$ transformation was used to construct a significance index for viability loss, where larger values indicate greater loss. Based on these analyses, we successfully screened five heat-tolerant lines (ranked by heat tolerance: CIMBL1 > CIMBL28 > GEMS36 > GEMS17 > R15X1141) and five sensitive control lines (ranked by sensitivity: TIE7922 > CIMBL141 > CML454 > 9642 > ZHENG28) (Fig. 8c). Genetic background analysis indicated that CIMBL1 originated from tropical/subtropical zones, consistent with the selective pressure exerted by high-temperature environments on germplasm traits (Yang et al., 2011). Furthermore, CIMBL1 displayed drought-resistant phenotypes at the seedling stage, whereas TIE7922 showed sensitivity (Liu, 2013), suggesting a possible association between pollen heat tolerance and drought resistance in maize.

To further explore genetic loci associated with pollen viability under high temperature stress, we used the ratio of pollen viability at 40 °C to that at 28 °C as the phenotype trait for analysis (Fig. 8d-8e). A GWAS was performed using 553,906 single nucleotide polymorphisms (SNPs) with a minor allele frequency (MAF) ≥ 0.05 (Fu et al., 2013). After incorporating population structure (Q) and kinship (K) matrices into a mixed linear model (MLM) to control false positives, 29 significant SNPs were identified at a threshold of -log10 (p) ≥ 5.70. Based on the linkage disequilibrium (LD) decay distance of 50 kb in this population (Fu et al., 2013), SNPs within 100 kb were grouped into a single quantitative trait locus (QTL). This defined 17 QTLs (QTL1-QTL17), two of which contained 4, 3 and 2 significant SNPs, respectively, while the remaining 11 QTLs contained only one SNP. Within the 50 kb region upstream and downstream of these QTLs, we identified 72 candidate genes, 19 of which were directly targeted by at least one significant SNP. Using published expression data (Chen et al., 2014), we further prioritized two genes, Zm00001d007802 and Zm00001d032293, from these 19 candidates, as both showed high expressions in pollen, anther or tassel (Fig. 8f). Zm00001d007802 encodes a kinesin protein; its Arabidopsis ortholog TPX2 (targeting protein for Xklp2) encodes a microtubule-associated protein and showed markedly altered expression during pollen germination and tube growth (Wang et al., 2008). Zm00001d032293 encodes a WD40 repeat-like superfamily protein, and its Arabidopsis ortholog has been linked to drought tolerance (Endo et al., 2008). Base on the significant SNPs within these two genes, the 350 lines were classified into two haplotype groups. Lines carrying haplotype 2 (Hap2) were fewer in number but exhibited higher pollen viability than those carrying Hap1 (Fig. 8g-8h),

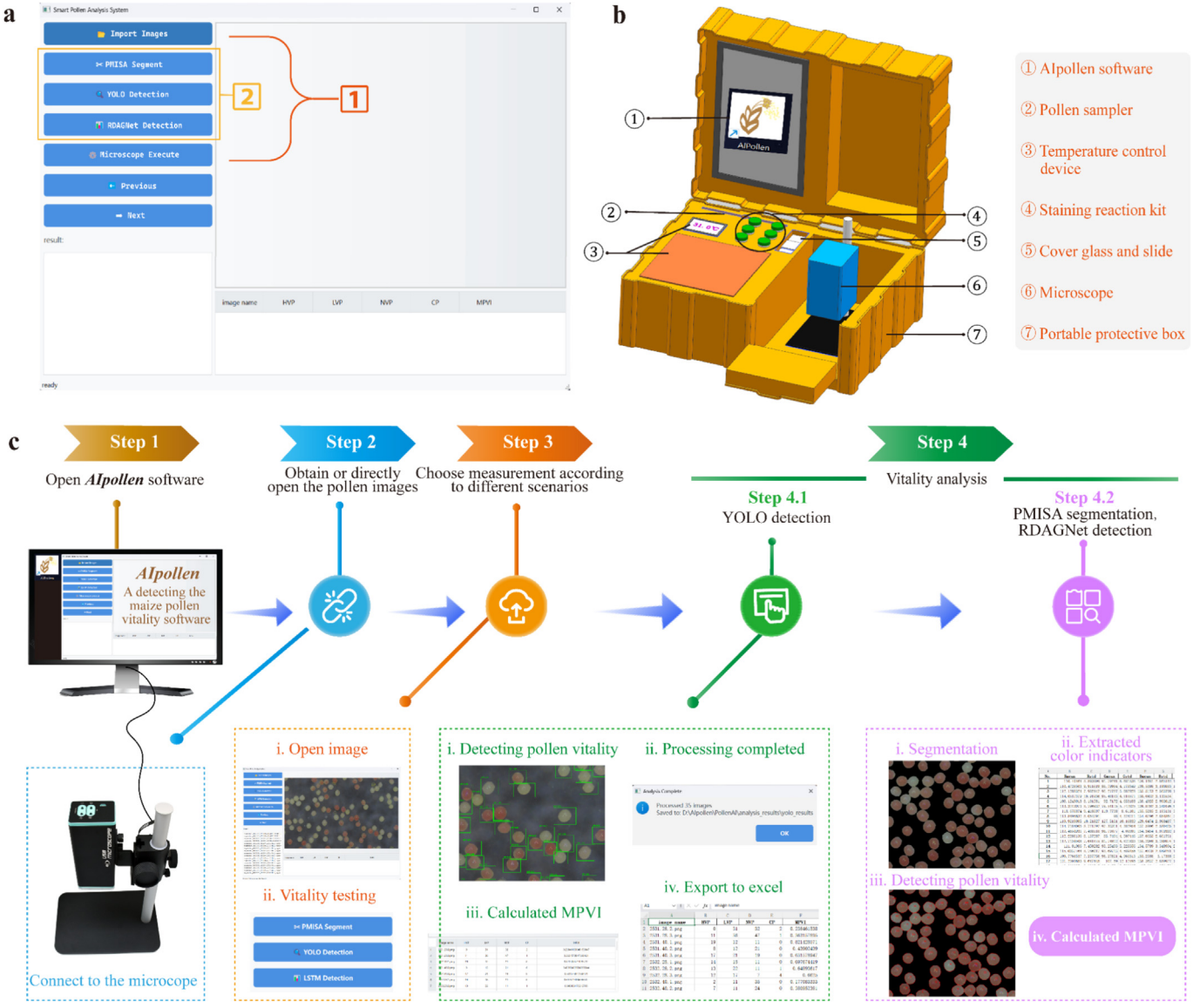


Fig. 7. Development of AIpollen software and PVDT device. (a) Graphical user interface of AIpollen; (b) Structure of the PVDT device; (c) Schematic workflow executed by AIpollen during microscopic pollen image analysis. Note: Step 4.1 represents pollen viability detection for normal density images, while step 4.2 focuses on high-density images. The results for a single image or all images in a folder are automatically saved in a subfolder named "Feature extract" under the "Results" directory. Saved outputs include detection, segmentation, numbered pollen images, and corresponding Excel files.

suggesting the potential utility of these haplotypes in breeding for pollen heat tolerance variety.

3.5.2. Field application of PVDT in evaluating pollen viability across species

To evaluate the performance of the PVDT tool in detecting pollen viability across different species, we collected pollen samples from ten plant species representing eight distinct families: wheat (*Poaceae*), scallion (*Amaryllidaceae*), pea (*Fabaceae*), alfalfa (*Fabaceae*), potato (*Solanaceae*), tobacco (*Solanaceae*), cucumber (*Cucurbitaceae*), honeysuckle (*Caprifoliaceae*), *salvia miltiorrhiza* (*Lamiaceae*), and rose (*Rosaceae*). We firstly analyzed key morphological parameters, LWR, Area and Roundness, to precisely compare differences among species. The ANOVA and Duncan's MRT results revealed that wheat pollen showed morphological similarities to maize pollen, particular in LWR, though maize pollen grains are rounder and larger (Supplementary Fig. 7). Meanwhile, scallion and pea pollen display typically elliptical shape with the largest LWR values among all species tested. In contrast, honeysuckle pollen was unique due to its triangular-circular appearance and significantly larger area compared to most other species. Potato, scallion, tobacco, rose, and alfalfa all had relatively small pollen areas, with potato pollen exhibiting the smallest overall area.

We stained the pollen samples using the TTC method. Three staining images per species were annotated using Labelme software and then used for training the YOLOv10-CBAM model via transfer learning techniques (Gholizade et al., 2025). The transferred model was subsequently tested using ten stained images of each species. The results showed that the model performed optimally for wheat, honeysuckle, cucumber and rose, achieving mAP@0.5 and accuracy exceeding 95% (Fig. 9). Notably, honeysuckle pollen viability yielded the best overall performance, with 98.3% accuracy and 97.3% mAP@0.5. On the other hand, detection accuracies for potato and tobacco were relatively poor (accuracy below 95% and mAP@0.5 below 90%). This lower performance is primarily attributed to the small

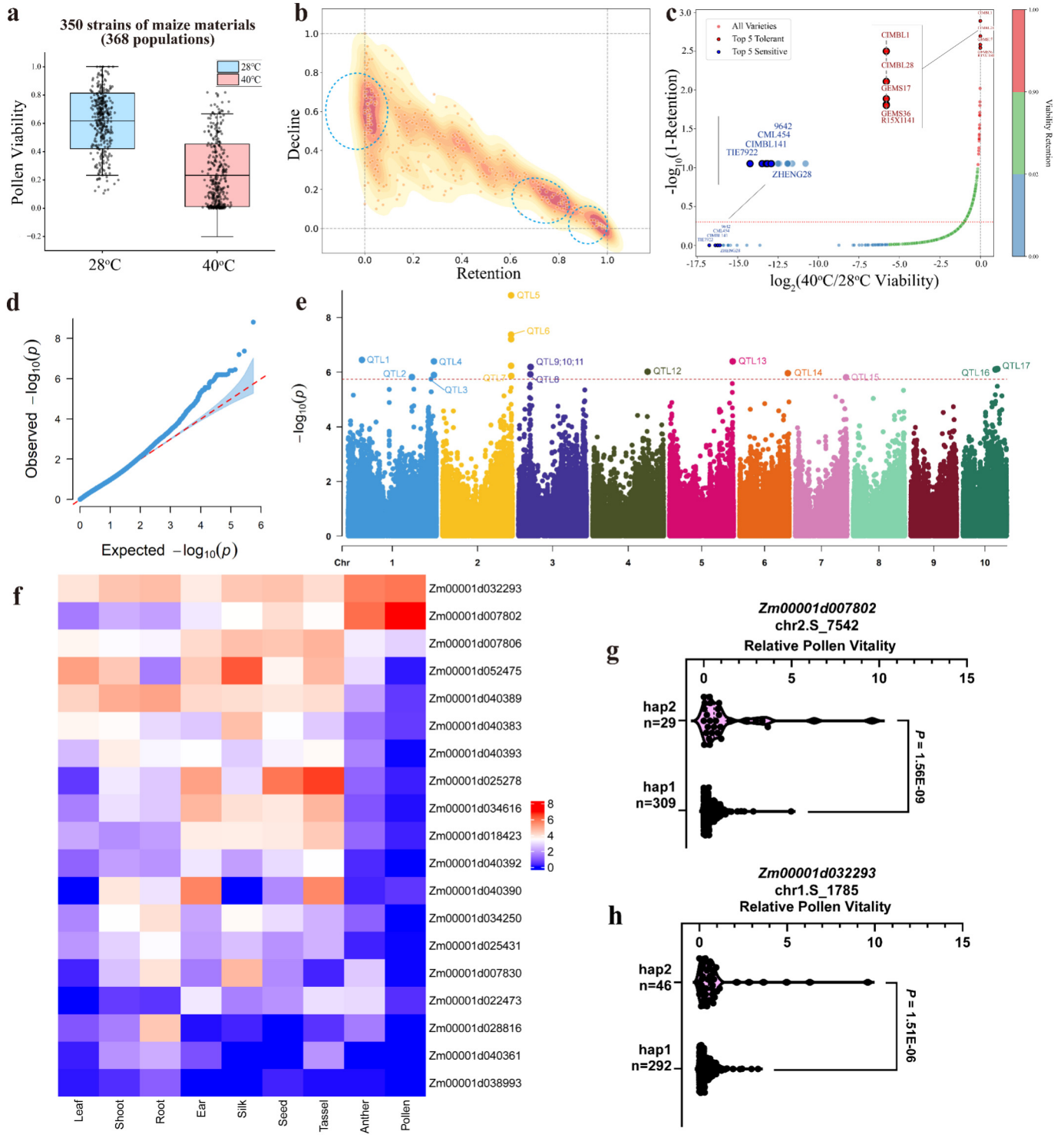


Fig. 8. Using PVDT to assess maize pollen viability for GWAS of viability related genes. (a) Pollen viability was evaluated by the PVDT at 28 °C and 40 °C across a maize association panel of 350 inbred lines. (b) Pollen viability declined values and retention rates across different inbred lines in the panel. (c) Screening of heat-tolerant and heat-sensitive inbred lines in the population. Red and blue dots represent tolerant and sensitive lines, respectively. (d-e) QQ plot (d) and Manhattan plot (e) of the GWAS for relative pollen viability at 40 °C relative to at 28 °C. The dashed horizontal line indicates the significance threshold ($p \leq 1$/effective SNP number (EN) = 1.81×10–6; where EN is 553,906). (f) Expression levels of the 12 candidate genes in different maize tissues based on expression data from the B73 line (Chen et al., 2014). (g-h) Comparison of relative pollen viability between different haplotypes of Zm00001d007802 (g) and Zm00001d032293 (h). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

size of these pollen grains, which complicates separation and segmentation. Alfalfa exhibited the lowest accuracy (91.5%) among all species tested.

Concerning error distribution, the primary detection errors for these pollens were concentrated in misjudgments between NVP and CP (Fig. 9a-9c). Critically, since both NVP and CP are classified as non-viable pollen, these misjudgments do not affect the overall accurate assessment of pollen viability. Together, the model's accuracy remained above 90% across all ten plant species, suggesting that the model, originally established for maize, can be effectively migrated and applied to

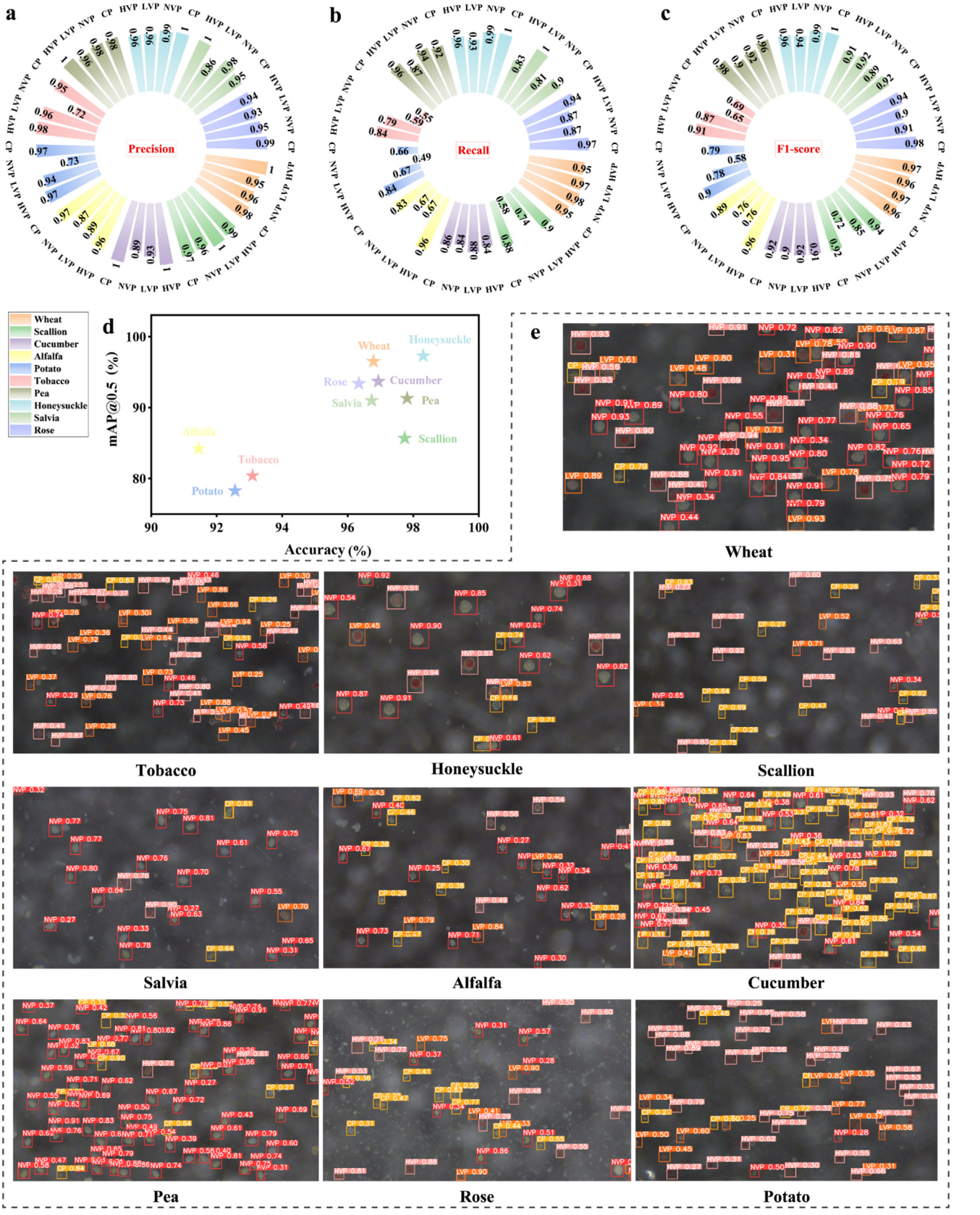


Fig. 9. Pollen viability detection of various plants based on transfer learning. (a-c) The precision, recall, and F1 score of pollen viability detection results for various plants. (d) The accuracy and mAP@0.5 of pollen viability detection results for various plants. (e) Identification images of pollen viability in various plants.

viability detection in diverse plant pollens using transfer learning techniques.

4. Discussion

4.1. Notable advantages of deep learning algorithms in pollen viability detection

The YOLOv10-CBAM (accuracy = 96.9%) and YOLOv10 (accuracy = 95.9%) models significantly outperformed the YOLOv8 (accuracy = 89.4%) in the evaluation of the Test1 dataset (Table 1). This is primarily attributed from YOLOv10's utilization of an enhanced version of Cross Stage Partial Network (CSPNet) as its backbone, which improves gradient flow and reduces computational redundancy, further enhances model performance (Wang et al., 2020). Further analysis revealed that the YOLOv10-CBAM model exhibited stronger generalization capabilities on Test2 datasets, including different lighting, pollen density, etc. (Fig. 5). This is primarily due to CBAM's ability to adaptively learn the importance of input feature maps, thereby enhancing the expressive capability of the features (Zhang et al., 2025). When building pollen classification models based on PMISA segmentation for images, the accuracy of deep learning models (CNN: 87.9%; RDAGNet: 97.9%) outperform traditional machine learning models (OEL: 82.7%; SVM: 83.0%; XGBoost:84.4%; MLP: 85.8%) (Table 2). This advantage comes from deep learning's strong feature extraction and representation learning abilities, which enable automatic learning of high-level features from raw data (Dhanya et al., 2022; Ritharson et al., 2024). Notably, RDAGNet achieved the highest performance, surpassing traditional CNN architectures by a 10% improvement. This advantage primarily originates from the synergistic integration of the residual network framework and the dual-layer attention mechanism. Specifically, residual connections mitigate gradient vanishing issues and enable effective deep feature learning through skip connections, while the dual-layer attention mechanism enhances feature representation quality. This combined architecture allows RDAGNet to capture discriminative patterns in color-texture sequences more effectively than single-modality attention mechanisms.

4.2. Significant impact of pollen density and image quality in pollen viability detection

The results of pollen detection based on microscopic images indicate that the pollen datasets of Test2-HD and Test2-LQ have the most significant impact on the model's accuracy. The possible reason is that in high-density pollen images, pollen grains are tightly packed, significantly increasing the difficulty for the model to identify them, thereby leading to a decline in both the accuracy and detection rate of the model. This possibility was supported by a previous research, where they found that the density of tea buds in images can significantly affect the performance of detection models (Gui et al., 2024). In light of this, it is recommended that during pollen data collection, pollen density should be appropriately reduced to enhance the accuracy of detection models. Concerning to image quality, low quality image makes the originally clear details becoming blurred, and colors deviating from their original values. These changes increase the difficulty for the model to correctly identify the content of the image (Albert et al., 2023). Loss of transparency may cause the model to fail to accurately distinguish objects from their backgrounds, which explains the decrease in the model's detection. Therefore, maintaining high-quality pollen images and reducing pollen density are crucial to ensure that the model can accurately and reliably detect pollen.

4.3. Significantly enhances PMISA in pollen morphology preservation and detection accuracy

High-density and low-quality pollen datasets are the primary factors affecting the recall and accuracy of models. PMISA effectively addresses this issue by achieving a pollen detection of 99.8% (Supplementary Table 9), which is significantly superior to the YOLOv10-CBAM (Table 1). More importantly, during the segmentation process, the PMISA algorithm ensures the maximization of pollen morphology features preservation, which is a capability lacking in the YOLO series models. By introducing the circular hough transform for circle detection to intelligently distinguish and eliminate non-circular impurities, effectively purifying the detection environment. This technique is particularly suitable for the extraction of circular targets, such as red blood cell counting (Aslani et al., 2022) and pupil recognition (Huang et al., 2023). Meanwhile, the combination of active contour model and morphological processing can optimize the true contour of pollen grains, improving the smoothness and naturalness of the contour. Additionally, the watershed algorithm can further separate touching pollen grains, solving the problem of high pollen density. In subsequent accuracy assessments based on models built using the segmented Test2-HD and Test2-LQ datasets, the RDAGNet model achieved an accuracy of 97.9% and 97.1%, respectively, representing an approximately 7% improvement compared to the previous YOLOv10-CBAM accuracy of 91.5% and 89.4%. Furthermore, PMISA can also achieve background removal and pollen detection in complex backgrounds, which endows AIpollen software with higher flexibility and practicality in practical applications, providing a second solution for pollen viability detection.

4.4. Low-cost and high-efficiency of PVDT offer a fieldwork tool for breeders

An assessment of the efficiency of PVDT in pollen viability detection revealed that it can achieve accurate recognition within 0.5 s when processing high-density pollen images. In comparison, manual identification methods require 3 to 5 min, indicating that PVDT is over 360 times as efficient. Currently, the rapid pollen viability detection technologies adopted globally primarily rely on flow cytometers. These devices, such as the Ampha Z40 (Amphasys, Switzerland) and Cube6 V2m (Sysmex, Japan), assess pollen viability by collecting and analyzing scattered light signals and fluorescence signals from pollen. While flow cytometers excel in detection speed and accuracy, their high cost is a significant disadvantage. Pollen viability analyzers integrated with flow cytometry usually exceed 600,000 CNY in retail price. Nevertheless, our developed PVDT system is available at around 50,000 CNY, representing a 12–15-fold reduction in cost compared to mainstream commercial instruments. With this equipment, we assessed pollen viability for 350 maize inbred lines, and identified *Zm00001d007802 and Zm00001d032293* candidate genes with natural allele responding for diverse pollen viability among different lines. Further functional validation of these genes might provide new genetic resources for breeding of maize variety with high pollen viability. Together, this equipment not only makes pollen viability detection more affordable but also provides breeders with a cost-effective and efficient aid for fieldwork.

CRediT authorship contribution statement

He Li: Writing – review & editing, Writing – original draft, Validation, Methodology. **Xuemei Du:** Writing – original draft, Validation. **Yilin Mao:** Writing – original draft, Methodology, Investigation. **Han Zhang:** Software. **Hongtao Shi:** Methodology. **Xinning Li:** Software. **Juhong Zhang:** Investigation. **Zihui Wang:** Investigation. **Riliang Gu:** Writing – review & editing, Project administration, Funding acquisition. **Qun Sun:** Writing – review & editing, Project administration, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. We declare that we have not used any AI-generated content (AIGC) tools.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aiia.2026.05.011.

Data availability

The datasets are available at https://github.com/HekeLi/pollen.

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